The suspension-loop adjunction: pointed maps out of the suspension are pointed maps into the loop space

The Univalent Foundations Program

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Editorial note. The counted statement is the author's ‘lem:susp-loop-adj‘ (source lines 678-682, Lemma 6.5.4); the proof environment begins on line 684, the line after \end{lem} apart from the counter-adjustment line 683, so the pairing is by adjacency.

Editorial note. Source line 683, ‘\addtocounter{thm}{1}‘ with its comment about a removed numbered equation, is a numbering adjustment internal to the pinned source and is not part of the proof; it is excluded from the retained range, and the printed number is restored by initial_counters instead.

Editorial note. The retained context defines the loop space and the suspension action on pointed types, which the statement quantifies over.

Editorial note. The retained counter values (chapter 6, section 5, theorem offset 3) make the retained lemma print as Lemma 6.5.4, matching the original book numbering.

Editorial note. The four external references are navigation pointers to the suspension and sigma-type facts the retained proof invokes by label; the counted proof is retained in full and nothing is replaced by them.

Editorial scope. One original result of the HoTT Book, counted once: the suspension-loop adjunction Map_*(susp A, B) ~ Map_*(A, Omega B) (Lemma 6.5.4).

Allowed standard prerequisites (editorial).

- Universal property of suspensions. susp A -> B is equivalent to the type of triples (b_n, b_s, A -> (b_n = b_s)), with the map from right to left given by the suspension recursor.. It produces the first equivalence of the chain at source lines 687-688 and is invoked again at source lines 693-695.
- Associativity and commutation of sigma types. Sigma types reassociate and their components may be reordered without changing the type up to equivalence.. Source lines 689-691 and 703-704 are consecutive reassociations of a four-component sigma type.
- Contractibility of the based path space. For any b_0 : B the type (Sigma b : B, b = b_0) is contractible with centre (b_0, refl b_0).. It removes that component at source line 697 and again at source line 707.

Source reference. Original exercise 6.11: sources/hott-hits/pinned-original-source.tex, source lines 2174–2174. Pointer to the exercise where the universal property of suspensions used at source lines 693-695 is stated. It is a navigation pointer only: the author's proof of the counted lemma is retained line for line and is complete without the exercise.

Source reference. Original exercise 2.10: sources/hott-hits/book-labelnumbers-first-edition.tex, source lines 1402–1402. Original result cited by the retained text but located outside the retained span. It is a navigation pointer: the counted statement's own author proof is retained line for line, and no part of it is replaced by this reference.

Source reference. Original theorem 3.11.9: sources/hott-hits/book-labelnumbers-first-edition.tex, source lines 1844–1844. Original result cited by the retained text but located outside the retained span. It is a navigation pointer: the counted statement's own author proof is retained line for line, and no part of it is replaced by this reference.

Source reference. Original theorem 3.11.8: sources/hott-hits/book-labelnumbers-first-edition.tex, source lines 1760–1760. Original result cited by the retained text but located outside the retained span. It is a navigation pointer: the counted statement's own author proof is retained line for line, and no part of it is replaced by this reference.

In particular, note that $\mathbf{2} \simeq \mathbf{1}_+$. Thus, for any pointed type B we have

$$\mathsf{Map}_*(\mathbf{2}, B) \simeq (\mathbf{1} \to B) \simeq B.$$

Now recall that the loop space operation Ω acts on pointed types, with definition $\Omega(A, a_0) :\equiv (a_0 =_A a_0, \mathsf{refl}_{a_0})$. We can also make the suspension Σ act on pointed types, by $\Sigma(A, a_0) :\equiv (\Sigma A, \mathsf{N})$.

Lemma 6.5.4. *For pointed types (A, a_0) and (B, b_0) we have*

$$\mathsf{Map}_*(\Sigma A, B) \simeq \mathsf{Map}_*(A, \Omega B).$$

Proof. We first observe the following chain of equivalences:

$$\begin{aligned}
\mathsf{Map}_*(\Sigma A, B) &:\equiv \sum_{f:\Sigma A \to B} (f(\mathsf{N}) = b_0) \\
&\simeq \sum_{f:\sum_{(b_n:B)} \sum_{(b_s:B)} (A \to (b_n = b_s))} (\mathsf{pr}_1(f) = b_0) \\
&\simeq \sum_{(b_n:B)} \sum_{(b_s:B)} \big(A \to (b_n = b_s)\big) \times (b_n = b_0) \\
&\simeq \sum_{(p:\sum_{(b_n:B)} (b_n = b_0))} \sum_{(b_s:B)} (A \to (\mathsf{pr}_1(p) = b_s)) \\
&\simeq \sum_{b_s:B} (A \to (b_0 = b_s))
\end{aligned}$$

The first equivalence is by the universal property of suspensions, which says that

$$\Big(\Sigma A \to B\Big) \simeq \Big(\sum_{(b_n:B)} \sum_{(b_s:B)} (A \to (b_n = b_s))\Big)$$

with the function from right to left given by the recursor (see original exercise 6.11). The second and third equivalences are by original exercise 2.10, along with a reordering of components. Finally, the last equivalence follows from original theorem 3.11.9, since by original theorem 3.11.8, $\sum_{(b_n:B)}(b_n = b_0)$ is contractible with center $(b_0, \mathsf{refl}_{b_0})$.

The proof is now completed by the following chain of equivalences:

$$\begin{aligned}
\sum_{b_s:B}(A \to (b_0 = b_s)) &\simeq \sum_{(b_s:B)} \sum_{(g:A\to(b_0=b_s))} \sum_{(q:b_0=b_s)} (g(a_0) = q) \\
&\simeq \sum_{(r:\sum_{(b_s:B)}(b_0=b_s))} \sum_{(g:A\to(b_0=\mathsf{pr}_1(r)))} (g(a_0) = \mathsf{pr}_2(r)) \\
&\simeq \sum_{g:A\to(b_0=b_0)} (g(a_0) = \mathsf{refl}_{b_0}) \\
&\equiv \mathsf{Map}_*(A, \Omega B).
\end{aligned}$$

Similar to before, the first and last equivalences are by original theorems 3.11.9 and 3.11.8, and the second is by original exercise 2.10 and reordering of components. □